

Materials for the Moon: Developing Physically Based Lunar Regolith Assets to Support Space Training and Simulation

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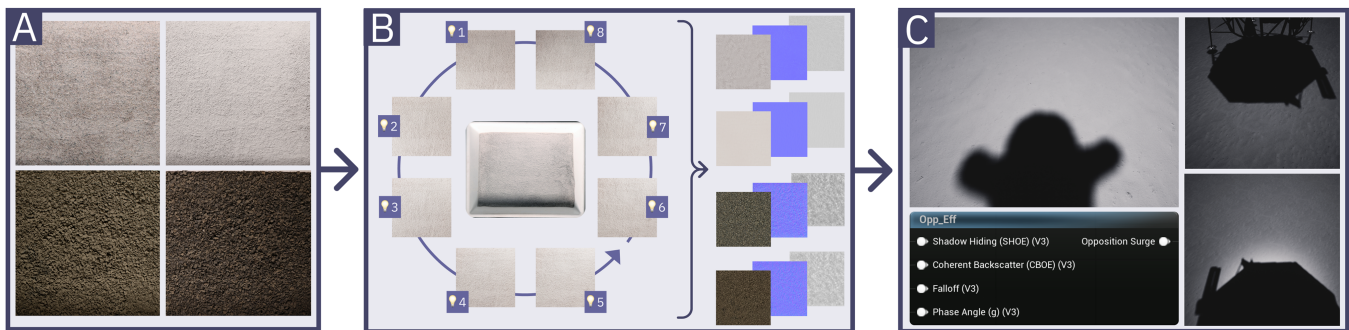


Figure 1: Producing physically based materials for the lunar surface: (A) regolith simulant samples representing the Moon’s main terrain types, (B) photometric stereo capture and map generation, (C) creating custom Unreal Engine materials reproducing lunar regolith’s photometric properties and rendering in real time - validated throughout by planetary science experts.

Abstract

As humanity prepares to return to the Moon for the first time in over fifty years and with numerous robotic missions on the horizon, XR training and photorealistic robotics simulators are becoming essential for lunar exploration. However, these tools are held back by inaccurate representation of the regolith that covers the lunar surface. To help close this realism gap, we present work towards a free library of high-fidelity physically based lunar regolith textures and materials to support the success of future missions to the Moon.

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1 Introduction

Humanity is preparing to return to the Moon for the first time in over fifty years, with NASA and the China Manned Space Agency

planning crewed landings by the end of the decade, alongside numerous robotic missions from national space agencies and private companies set to launch in the coming years.

In this context, game engines such as Unreal Engine (UE) and other interactive real-time 3D platforms are becoming essential tools for space exploration. Extended reality (XR) training applications allow astronauts to rehearse scenarios and experience environmental conditions impossible to reproduce on Earth [Bingham et al. 2025]. Advances in machine learning have spurred interest in using physically based, photorealistic simulations to train autonomous systems to traverse the lunar terrain [Richard et al. 2024].

However, these tools are often held back by their inaccurate representation of lunar regolith—the thick layer of fine, fragmented particles of rock, glass, and minerals that covers the Moon’s surface—and its unusual photometric properties which pose challenges for both human and machine perception, impacting mission safety and success. Despite the growing use of interactive real-time 3D simulation in the space sector, there remains a pronounced lack of free, high-quality, physically based materials and 3D assets tailored to the Moon’s surface, leading developers to resort to generic substitutes such as gravel or ash, which, while superficially plausible, do not reflect the appearance and physical behaviour of lunar regolith [Richard et al. 2024]. These inaccuracies can not only hinder accurate identification of geological features for sample return missions, but also prevent astronauts and robotic assets from developing the capabilities necessary to safely and effectively explore the lunar environment.

To help close this realism gap, we present preliminary results from a project in collaboration with the European Space Agency

*Work performed during internship at the European Space Agency, January–July 2025.

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(ESA) and planetary scientists in its VULCAN regolith simulant laboratory. Our aim is to create the first freely-available library of physically based lunar regolith materials based on real imagery, scientific data, and expert validation, ready for deployment in XR training, simulation, and science communication.

2 Approach

Our approach has two main components: first, capturing high-fidelity surface textures from regolith simulants —materials designed to replicate the properties of lunar regolith for research and testing—, and second, developing custom Unreal Engine materials that reproduce lunar regolith’s unique photometric behaviors in real time. Throughout, we worked with planetary scientists expert in regolith simulant development and characterisation who selected samples, provided scientific guidance, and validated our results.

Texture Capture and Calibration. Lunar regolith differs significantly from terrestrial soils in both composition and appearance. Since returning regolith samples to Earth is a key objective of future missions to the Moon, accurately rendering its unique geology is of vital importance for training astronauts and vision-based systems.

We captured four carefully prepared regolith simulant samples representing the Moon’s main terrain types using photometric stereo. Two samples modelled the lighter, anorthositic Highlands that cover ~83% of the Moon’s surface and two the darker, basaltic Maria that make up the remaining ~17%.

We derived surface normals and albedo from the captured imagery, and generated roughness maps, calibrating all values against data from the scholarly literature (e.g., remote sensing, analyses of Apollo samples) and with input from experts. We then processed the validated textures using procedural generation techniques to create collections of tiling texture maps for each simulant sample, ready for integration into Unreal Engine and other real-time 3D environments.

Photometric Behaviour and UE Material Implementation. Lunar regolith is characterised by low albedo, high cohesiveness and porosity due to low gravity, and non-Lambertian reflectance with a strong backward scattering tendency [Hapke 2012]. As a result, the appearance of the lunar surface varies significantly depending on Sun angle and observer position, creating perceptual challenges that impaired situational awareness for Apollo astronauts [Rahill 2019] and which remain a key concern for modern optical navigation systems. This underscores the importance of accurately reproducing the photometric behaviours of lunar regolith to prepare crews and robotic systems for future lunar missions.

Among these photometric behaviours, the most striking phenomenon is the Opposition Effect (OE): a pronounced surge in brightness at very low phase angles, when the Sun is directly behind the observer. The OE arises from coherent backscatter and shadow hiding at the particle scale [Hapke 2012]. Crucially, this brightness peak can ‘wash out’ textural cues essential for depth and shape estimation [Rahill 2019].

To replicate these complex phenomena, we developed custom materials for each terrain type in Unreal Engine —ESA’s preferred platform— capable of ingesting the captured textures and approximating lunar regolith’s unique properties based on Hapke’s [2012]

photometric model, the most widely used in planetary science. We leveraged UE’s Substrate material framework to model the effects of surface roughness and subsurface scattering in the simulated regolith. As UE’s BSDF does not account for the Opposition Effect, we implemented custom Substrate shader nodes to model it, successfully reproducing the low-phase brightness surge (Figure 2). Once again, we validated real-time rendered outputs with experts.

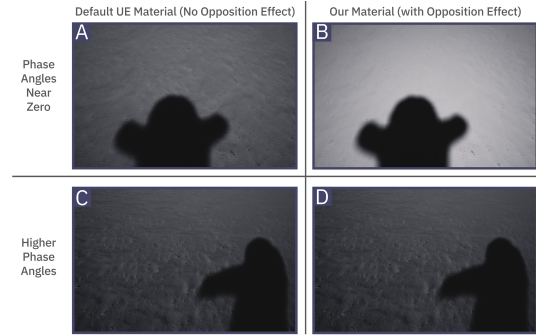


Figure 2: Testing the Opposition Effect under varying lighting conditions: at low phase angles, UE’s default material lacks OE (A) while our material correctly reproduces the brightness surge (B). At higher phase angles (C, D), there is no surge.

3 Discussion and Future Work

We propose a workflow for capturing and rendering lunar regolith in real time, grounded in real imagery, scientific data and expert input. The regolith textures are interoperable with any PBR environment. Initial performance tests in UE indicate the materials render at XR frame rates. Our solution does not rely on modifying UE’s source code, enabling easy integration into existing projects.

The purpose-built regolith materials exhibit markedly greater fidelity than gravel or ash stand-ins currently used at ESA. Our implementation of the Opposition Effect is, to our knowledge, the first approximation of this phenomenon in UE. The materials are already being integrated into XR projects at ESA, and internal feedback highlights the value of the enhanced fidelity this work brings to space training and simulation.

We aim to expand our material library beyond the four initial samples to include key sites such as Shackleton Crater, enhance our materials to account for the photometric effects of anthropogenic disturbances (e.g., footprints, rover tracks, blast zones), and evaluate their impact in real-world use cases ahead of a wider release.

This is a critical first step towards a freely available library of high-fidelity physically based lunar regolith materials to support the success of future human and robotic missions to the Moon.

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